Cody Freitag

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Professional Appointments

Cornell Tech, *Postdoctoral Researcher* Oct. 2026 – Present
- Host: Rafael Pass

Columbia University, *Adjunct Assistant Professor of CS* Jan. 2027 – May 2027
- Upcoming appointment

Proof Trading, *Research Consultant* Oct. 2026 – Dec. 2026 (expected)

Hebrew University of Jerusalem, *Postdoctoral Researcher* Sep. 2025 – Aug. 2026
- Host: Ilan Komargodski

Northeastern University, *Postdoctoral Researcher* Sep. 2023 – Aug. 2026
- Host: Daniel Wichs

Boston University, *Postdoctoral Researcher* June 2023 – Aug. 2023
- Host: Ran Canetti

Education

Cornell University Aug. 2017 – May 2023
- PhD in Computer Science
- MS in Computer Science
- Advisor: Rafael Pass

The University of Texas at Austin Aug. 2013 – May 2017
- BS in Computer Science, Turing Scholars Honors
- BS in Mathematics, Dean's Scholars Honors

Teaching

Instructor

Algorithms, *Columbia University* Spring 2027

Teaching Assistant

Networks and Markets, *Cornell University* 2019–2021 (4 semesters)

Algorithms and Complexity, *UT Austin* Spring 2016

Discrete Mathematics, *UT Austin* Fall 2015, Spring 2016

Computer Fluency, *UT Austin* Fall 2014, Spring 2015

Additional Teaching Programs

New Horizons in TCS — Summer 2021
- Teaching Assistant

Stanford Code in Place — Summer 2020
- Section Leader

Additional Experience

NTT Research, *Research Intern* — Fall 2021 – Spring 2022
- Host: Ilan Komargodski

Yeletech Security / Bolt Labs, *Cryptography Consultant* — June 2018 – Dec. 2018

Zeutro, *Software Developer* — June 2016 – Mar. 2018

Rutgers University, *Undergraduate Researcher* — Summer 2016
- DIMACS REU
- Host: Muthu Muthukrishnan

University of Maryland, *Undergraduate Researcher* — Summer 2015
- REU CAAR
- Host: Jonathan Katz

Bloomberg LP, *Software R&D Intern* — Summer 2014

L-3 Communications, *Software Engineering Intern* — Summer 2013

Honors and Awards

- Khoury College Distinguished Postdoctoral Fellowship — Sep. 2023 – Aug. 2025
- NSF Graduate Research Fellowship — June 2019 – May 2021, June 2022 – May 2023
- Cornell University Fellowship — Fall 2017 – Spring 2018
- UT Computer Science Best Undergraduate Thesis Award — Spring 2017
- UT Unrestricted Endowed Presidential Scholarship — Fall 2016 – Spring 2017
- UT CNS Distinguished College Scholar — Fall 2014 – Spring 2017

Service

- Program committee member for Crypto 2024, TCC 2025, TCC 2026
- Subreviewer for various conferences including STOC, FOCS, SODA, Crypto, Eurocrypt, TCC, Asiacrypt, ITCS, ICALP, PKC, SCN, DISC, SOSA
- Co-organizer of Cornell Cryptography Seminar — 2018 – 2021
- Cornell CS PhD admissions committee — 2020
- Volunteer for Cornell CS PhD admissions committee — 2018, 2019
- Cornell CS visit day leader — 2019

Publications

Conference Papers

- **Crypto 2026** *Unique SNARGs with Adaptive Security: Constructions and Black-Box Separations*
  Cody Freitag, Daniel Wichs
- **ITCS 2026** *Improved Rate for Non-Malleable Codes and Time-Lock Puzzles*
  Cody Freitag, Manu Kondapaneni, Ilan Komargodski, Jad Silbak
- **TCC 2025** *Seedless Condensers for Efficiently Samplable Sources*
  Cody Freitag, Jad Silbak, Daniel Wichs
- **STOC 2025** *Unambiguous SNARGs for P from LWE with Applications to PPAD Hardness*
  Liyan Chen, Cody Freitag, Zhengzhong Jin, Daniel Wichs
- **Eurocrypt 2024** *Public-Coin, Complexity-Preserving, Succinct Arguments of Knowledge for NP from Collision-Resistance*
  Cody Freitag, Omer Paneth, Rafael Pass
- **CCS 2023** *Riggs: Decentralized Sealed-Bid Auctions*
  Nirvan Tyagi, Arasu Arun, Cody Freitag, Riad Wahby, Joseph Bonneau, David Mazières
- **Crypto 2023** *How to Use (Plain) Witness Encryption: Registered ABE, Flexible Broadcast, and More*
  Cody Freitag, Brent Waters, David J. Wu
- **ITC 2023** *The Cost of Statistical Security in Proofs for Repeated Squaring*
  Cody Freitag, Ilan Komargodski
- **Eurocrypt 2023** *Optimal Security for Keyed Hash Functions: Avoiding Time-Space Tradeoffs*
  Cody Freitag, Ashrujit Ghoshal, Ilan Komargodski
- **TCC 2022** *Parallelizable Delegation from LWE*
  Cody Freitag, Rafael Pass, Naomi Sirkin
- **TCC 2022** *Cosmic Security: Security Relative to Stateful Natures*
  Benjamin Chan, Cody Freitag, Rafael Pass
- **Crypto 2022** *Time-Space Tradeoffs for Sponge Hashing: Attacks and Limitations for Short Collisions*
  Cody Freitag, Ashrujit Ghoshal, Ilan Komargodski
- **TCC 2021** *Non-Malleable Time-Lock Puzzles and Applications*
  Cody Freitag, Ilan Komargodski, Rafael Pass, Naomi Sirkin
- **SCN 2020** *Impossibility of Strong KDM Security with Auxiliary Input*
  Cody Freitag, Ilan Komargodski, Rafael Pass
- **Eurocrypt 2020** *SPARKs: Succinct Parallelizable Arguments of Knowledge*
  Cody Freitag, Ilan Komargodski, Rafael Pass, Naomi Sirkin
- **Eurocrypt 2020** *Continuous Verifiable Delay Functions*

Cody Freitag, Ilan Komargodski, Rafael Pass, Naomi Sirkin

- **Crypto 2019** *Non-uniformly Sound Certificates with Applications to Concurrent Zero-Knowledge*
  Cody Freitag, Ilan Komargodski, Rafael Pass
- **AISTATS 2019** *Test without Trust: Optimal Locally Private Distribution Testing*
  Jayadev Acharya, Clément L. Canonne, Cody Freitag, Himanshu Tyagi
- **RANDOM 2017** *Testing Hereditary Properties of Sequences*
  Cody Freitag, Eric Price, and William Swarthworth
- **ACNS 2017** *Signature Schemes with Randomized Verification*
  Cody Freitag, Rishab Goyal, Susan Hohenberger, Venkata Koppula, Eysa Lee, Tatsuaki Okamoto, Jordan Tran, and Brent Waters
- **CSCML 2017** *Symmetric-Key Broadcast Encryption: The Multi-Sender Case*
  Cody Freitag, Nathan Klein, and Jonathan Katz

Journal Papers

- **Journal of Cryptology 2025** *Time-Space Tradeoffs for Sponge Hashing: Attacks and Limitations for Short Collisions*
  Cody Freitag, Ashrujit Ghoshal, Ilan Komargodski
- **Journal of the ACM 2022** *SPARKs: Succinct Parallelizable Arguments of Knowledge*
  Cody Freitag, Ilan Komargodski, Rafael Pass, Naomi Sirkin
- **IEEE Journal on Special Areas in Information Theory: Privacy and Security of Information Systems 2021** *Inference under Information Constraints III: Local Privacy Constraints*
  Jayadev Acharya, Clément L. Canonne, Cody Freitag, Ziteng Sun, Himanshu Tyagi
- **Polyhedron 2014** *Modeling of Late 3d Transition Metal Metathesis of tert-Butoxide Complexes with Amines*
  Cody Freitag, Francisco Birk, William Ou, and Thomas Cundari
- **Journal of the American Chemical Society 2012** *Variable Pathways for Oxygen Atom Insertion into Metal-Carbon Bonds: The Case of $Cp^*W(O)_2(CH_2SiMe_3)$*
  Jiajun Mei, Kurtis Carsch, Cody Freitag, T. Brent Gunnoe, and Thomas Cundari

Other Manuscripts

- **PhD Dissertation 2023** *How to Provably Leverage Time in Cryptography*
  Cody Freitag (advised by Rafael Pass)
- **Undergraduate Thesis 2017** *Testing and Searching Pattern Avoiding Sequences*
  Cody Freitag (advised by Eric Price)

Presentations

Unique SNARGs with Adaptive Security: Constructions and Black-Box Separations

- Talk at Crypto 2026 Conference — Aug. 2026

Improved Rate for Non-Malleable Codes and Time-Lock Puzzles

- Talk at ITCS 2026 Conference — Jan. 2026

Seedless Condensers for Efficiently Samplable Sources

- Talk at TCC 2025 Conference — Dec. 2025

Unambiguous SNARGs for P from LWE with Applications to PPAD Hardness

- Talk at Charles River Crypto Day — Apr. 2025
- Talk at University of Washington Cryptography Reading Group — Feb. 2025

Public-Coin, Complexity-Preserving, Succinct Arguments for NP from Collision-Resistance

- Talk at Eurocrypt 2024 Conference — May 2024
- Talk at MIT Cryptography and Information Seminar — Dec. 2023

How to Use (Plain) Witness Encryption: Registered ABE, Flexible Broadcast, and More

- Talk at Crypto 2023 Conference — Aug. 2023
- Talk at Boston University Security Seminar — June 2023

How to Provably Leverage Time in Cryptography

- PhD Thesis Defense — Apr. 2023

Cosmic Security: Universal Reductions Relative to a Stateful Oracle

- Talk at Boston University Security Seminar — Dec. 2022

Parallelizable Delegation from LWE

- Talk at TCC 2022 Conference — Nov. 2022

The Cost of Statistical Security in Proofs for Repeated Squaring

- Talk at ITC 2023 Conference — June 2023
- Talk at UT Austin Crypto Reading Group — Oct. 2022
- Talk at MIT Cryptography and Information Seminar — Sept. 2022
- Talk at Cornell Theory Seminar — Sept. 2022
- Talk at NYU Crypto Reading Group — Sept. 2022

SPARKs: Succinct Parallelizable Arguments of Knowledge

- Talk at Eurocrypt 2020 Conference — May 2020
- Talk at Bar Ilan University Cyber Center Colloquium — May 2020
- Talk at Boston University Security Seminar — May 2020

Non-Uniformly Sound Certificates with Applications to Concurrent Zero-Knowledge

- Talk at Crypto 2019 Conference — Aug. 2019
- Talk at Cornell Cryptography Seminar at Cornell Tech — Sept. 2019

Test without Trust: Optimal Locally Private Distribution Testing

- Poster at TPDP 2018 Workshop as part of CCS 2018 — Oct. 2018
- Talk at Cornell Cryptography Seminar at Cornell Tech — Oct. 2018

Testing and Searching Pattern Avoiding Sequences

- Talk at Cornell Theory Tea at Cornell — Mar. 2018
- Undergraduate Thesis Defense at UT Austin — May 2017

Pan-Private Graph/Geometric Streaming Algorithms

- Talk at DIMACS REU at Rutgers University — July 2016

Symmetric-Key Broadcast Encryption: The Multi-Sender Case

- Talk at TSSA Undergraduate Research Talks at UT Austin — Nov. 2015
- Talk at DS ResULTS at UT Austin — Sept. 2015
- Talk at REU CAAR at UMD — Aug. 2015